

Asal Roudbari

Ph.D. Student in Machine Learning, Georgia Institute of Technology

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asalroudbari.github.io | [Google Scholar](https://scholar.google.com/citations?user=...) | github.com/asalroudbari | linkedin.com/in/asalroudbari

EDUCATION

Georgia Institute of Technology

Atlanta, GA

Ph.D. in Machine Learning, [School of Industrial and Systems Engineering \(ISyE\)](#) 2024 – Present (expected 2028)

- Advisor: [Prof. Pascal Van Hentenryck](#), NSF AI Institute for Advances in Optimization (AI4OPT). GPA: 3.7/4.0
- Research focus: representation learning for irregularly sampled multivariate time series, survival analysis and time-to-event modeling, interpretable temporal clustering, uncertainty quantification, with applications in healthcare
- Coursework: Sequence Prediction, Computational Statistics, Conversational AI (in progress)

Sharif University of Technology

Tehran, Iran

B.Sc. in Electrical Engineering (Control and Dynamical Systems)

2020 – 2024

SKILLS

- **Languages and frameworks:** Python, PyTorch, scikit-learn, NumPy, Pandas, OpenCV, R, MATLAB, LaTeX
- **Machine learning:** time-series representation learning, deep sequence models, contrastive learning, clustering, graph neural networks, time-series and tabular foundation models, online learning
- **Health and clinical ML:** survival analysis, time-to-event modeling with censored outcomes, longitudinal clinical and EHR data, biomarker and radiographic data, model interpretability (SHAP), collaboration with clinicians and surgeons

PUBLICATIONS

Conferences

- [1] **A. Roudbari**, P. Van Hentenryck, and R. Zandehshahvar. “Interpretable Phenotype Discovery Using Temporal Risk-Aware Clustering.” *NeurIPS 2026 Workshop on Interpretability4Discovery* (accepted).
- [2] **A. Roudbari**, M. Dalvand, N. Pasiar, N. Asadi, M. Yousefi, and M. T. Masouleh. “Autonomous Vision-Based Robotic Grasping of Household Objects: A Practical Case Study.” *11th RSI International Conference on Robotics and Mechatronics (ICRoM)*, IEEE, 2023. [[IEEE Xplore](#)]

Clinical Conferences

- [3] M. C. Welborn, **A. Roudbari**, R. Zandehshahvar, P. Van Hentenryck, et al. “Collagen X Biomarker (CXM) is Predictive of Growth Cessation in Idiopathic Scoliosis.” *Scoliosis Research Society (SRS) Annual Meeting*, 2025. **John H. Moe Award for Best Basic Science/Translational Research Poster**. [[SRS announcement](#)]
- [4] M. C. Welborn, **A. Roudbari**, R. Zandehshahvar, J. O. Sanders, P. Van Hentenryck, et al. “Biomarker Longitudinal Idiopathic Scoliosis Study: The Impact of Growth on Curve Progression in Braced Patients with Idiopathic Scoliosis Who Experience Curve Progression vs. Curve Regression.” *Pediatric Orthopaedic Society of North America (POSNA) Annual Meeting*, 2025. **Nominated for Best Translational Research Oral Presentation**.

Journals

- [1] S. Sabzekar, **A. Roudbari**, A. Dehghani, A. Safaeiestalkhizir, and Z. Amini. “The Impact of Network Indices Integration on Traffic Flow Imputation Accuracy: A Machine Learning Approach.” *IEEE Transactions on Intelligent Transportation Systems*, 26(4):5411–5421, 2025. [[doi:10.1109/TITS.2025.3527019](https://doi.org/10.1109/TITS.2025.3527019)]
- [2] **A. Roudbari**, M. Khosravy, and O. Witkowski. “Genetic Algorithm Applications to Power Management.” In *Frontiers in Genetics Algorithm Theory and Applications*, Springer Nature, 2024. [[doi:10.1007/978-981-99-8107-6_9](https://doi.org/10.1007/978-981-99-8107-6_9)]

Under Review

- [3] **A. Roudbari**, M. C. Welborn, S. Sienko, BLISS Study Group, R. Zandehshahvar, and P. Van Hentenryck. “Personalized Prediction of Growth Cessation in Adolescents with Idiopathic Scoliosis Using Longitudinal Clinical Data and a Type X Collagen Biomarker: A Prospective Cohort Study.” Submitted to *Clinical Orthopaedics and Related Research (CORR)*.
- [4] **A. Roudbari**, R. Zandehshahvar, P. Van Hentenryck, M. C. Welborn, and S. Sienko. “Growth Markers and the Timing and Magnitude of Curve Change in Idiopathic Scoliosis.” Submitted to the *34th International Meeting on Advanced Spine Techniques (IMAST)*, 2027.

RESEARCH EXPERIENCE

Ph.D. Researcher, AI4OPT (NSF AI Institute for Advances in Optimization)

Sep 2024 – Present

Georgia Tech and Shriners Children's. Supervisors: [Dr. MohammadReza Zandehshahvar](#), [Prof. Pascal Van Hentenryck](#)

- **Interpretable temporal phenotyping (TRACE).** Designed a temporal clustering model for irregular multivariate time series that learns a time-aware representation of each patient's visit history, maps it to a soft phenotype that updates with every new visit, and predicts the future diagnosis only through that phenotype. On ADNI (1,788 participants, 10,941 visits) it gave the best phenotype separation at 24 months of five methods (silhouette index 42.0% vs. 36.2% for the next best). First-author paper accepted at the NeurIPS 2026 Interpretability4Discovery Workshop.
- **Time-to-event prediction for scoliosis.** Built a personalized pipeline that predicts time to growth cessation in adolescent idiopathic scoliosis from irregular longitudinal clinical, radiographic, and blood-biomarker (CXM) data, combining landmark survival analysis with recurrent sequence encoders and handling censored outcomes. Trained and validated on a multicenter prospective cohort of 300+ patients. First-author manuscript under review at CORR.
- **Curve progression under bracing.** Developed a landmark-anchored framework that predicts curve progression in braced patients from the CXM biomarker and height velocity; worked with orthopedic clinicians at Shriners Children's to align model outputs with decisions such as when to stop bracing. This work received the SRS John H. Moe Award.
- **Ongoing.** Extending TRACE to time-to-event outcomes (TRACE-Surv) and developing conformal calibration methods for uncertainty-aware survival prediction on irregular time series.

Undergraduate Researcher, Sharif University of Technology

Jul 2023 – Dec 2023

Tehran, Iran. Supervisor: [Prof. Zahra Amini](#)

- Developed a line graph neural network for spatial traffic flow imputation and studied how integrating network indices affects imputation accuracy; published in IEEE Transactions on Intelligent Transportation Systems.

Undergraduate Researcher (B.Sc. Project), University of Tehran

Apr 2023 – Sep 2023

Human and Robot Interaction Lab, Tehran, Iran. Supervisor: [Prof. Mehdi Tale Masouleh](#)

- Built an autonomous vision-based pipeline for grasping household objects with a 2-finger gripper on a Delta robot; first-author paper at ICROM 2023.

Research Collaborator, Cross Labs

Feb 2023 – Apr 2023

Remote, Japan. Supervisor: [Dr. Mahdi Khosravy](#)

- Wrote a book chapter on applications of genetic algorithms in power management (first author, Springer Nature).

Summer Research Intern, Marine Technology Group, Aalto University

Jul 2022 – Oct 2022

Finland. Supervisors: [Prof. Osiris Valdez Banda](#) and [Dr. Meriam Chaal](#)

- Studied wide-range estimation methods for object localization at sea on ECAMARIS, a project with ABB; worked with postdoctoral researcher [Ahmad BahooToroody](#).

HONORS AND AWARDS

All four are for the poster "Machine Learning for Personalized Risk Assessment in Idiopathic Scoliosis."

- **Poster Award, 2nd Place**, State of Science Meeting, Shriners Children's Jan 2026
- **Poster Award, 1st Place**, NSF AIVO Summit for AI Institutes Leadership Oct 2025
- **John H. Moe Award**, Best Basic Science/Translational Research Poster, SRS Annual Meeting [\[announcement\]](#) Sep 2025
- **Nominee**, Best Translational Research Oral Presentation, POSNA Annual Meeting Sep 2025

LEADERSHIP AND SERVICE

- **Student Leadership Member, AI4OPT** (Mar 2025 – Present). Represent PhD students; organizing *The Full Stack Researcher*, a ten-episode research communication workshop series across Georgia Tech, USC, and UC Berkeley.
- **President** (Oct 2025 – Mar 2026) and **Communications Director** (Jan 2025 – Oct 2025), **Iranian Student Association, Georgia Tech**. Directed all communication channels and coordinated cultural and academic programming.
- **Communications Director, Resana (EE Student Council), Sharif University of Technology** (May 2021 – Sep 2022). Organized seminars and workshops, co-organized the international competition "Spark" (I.D.E.A. Association, Germany), and hosted events with 300+ participants.